

Lecture Series 5A: Multiple Linear Regression CM2018

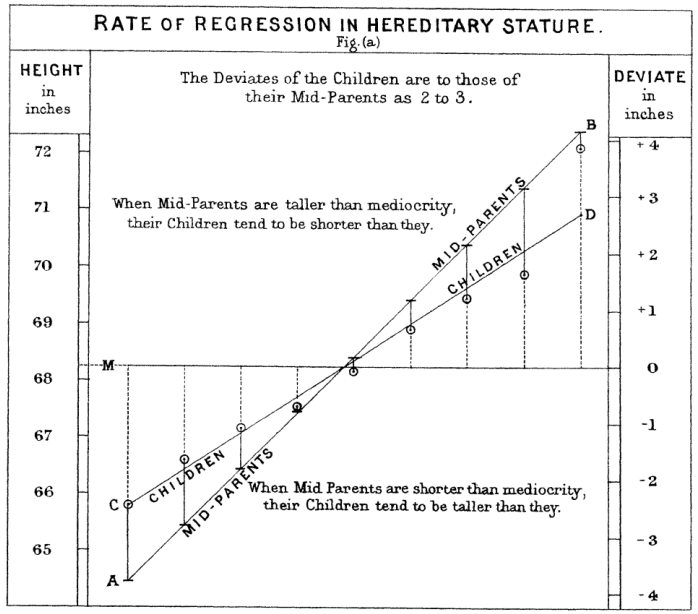
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L5-1 Multiple linear regression

Subset selection and the stepwise approaches

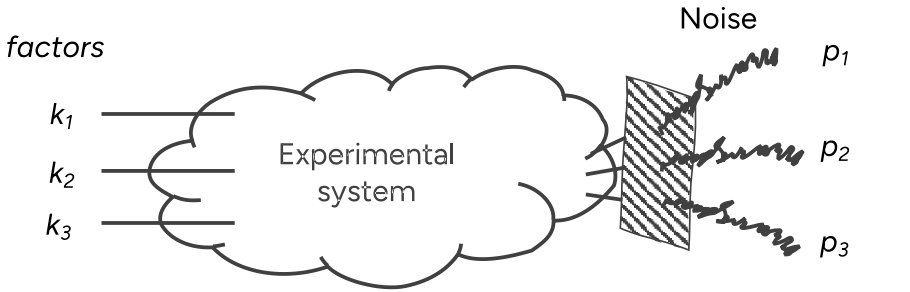
Multiple linear regression



$$\widehat{child.height} = \beta_0 + \beta_1 \cdot father.height + \beta_2 \cdot mother.height + \beta_3 \cdot grandparent.height + \cdots + \beta_j \cdot x_j$$

$H_0: \beta_1 = \beta_2 = \cdots = \beta_j = 0$

$H_1: \exists\, j \mid \beta_j \neq 0$



n entities = $k_3 \times p_3$

some model function of $y \leftarrow f(x)$

$y_1 \sim \beta_{1,1}\, x_1 + \mathcal{E}$

$y_2 \sim \beta_{2,1}\, x_1 + \beta_{2,3}\, x_3 + \mathcal{E}$

β coefficient

What is the impact of two or more independent variables, or their combined effect, on a dependent variable?

- Hypothesis testing.
- Explore and estimate causes.
- Predict effect.

Harrell 2015

Sanders 2019

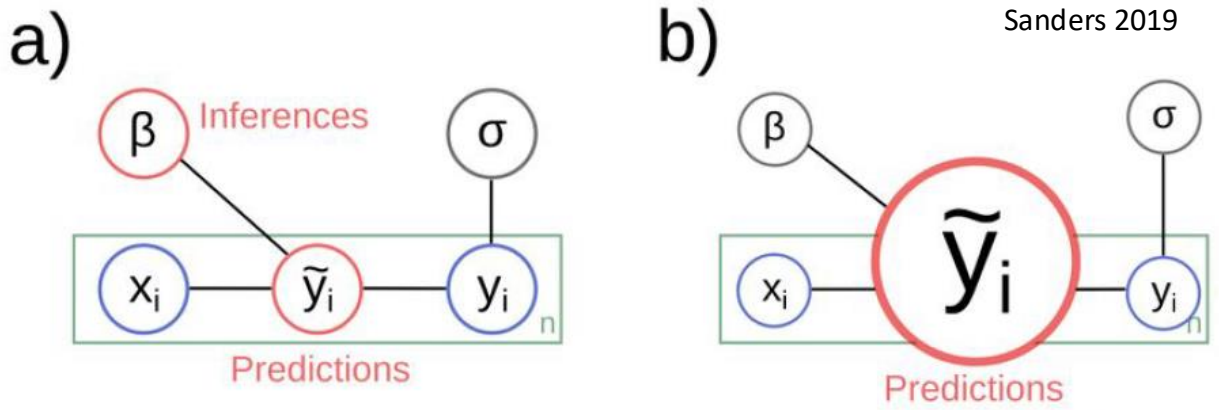


Figure 2. Illustration of perspectives on the sample model of Figure 1: a) a balanced perspective equally valuing inference and prediction and b) a prediction-oriented viewpoint.

Hypothesis testing

- Work with the full model, including insignificant variables.
- Carefully consider what interaction effects to consider and how these should be tested (*e.g.*, blocks of interactions or main effect and interaction effect).
- Consider the effect of imputing missing values.
- Consider the degrees of freedom to test the predictor of interest.
- Should not shrink parameter estimates of interest for hypothesis testing.
- Model validation is usually not necessary unless you wish to assess the degree of overfitting.

Harrell 2015

Prediction

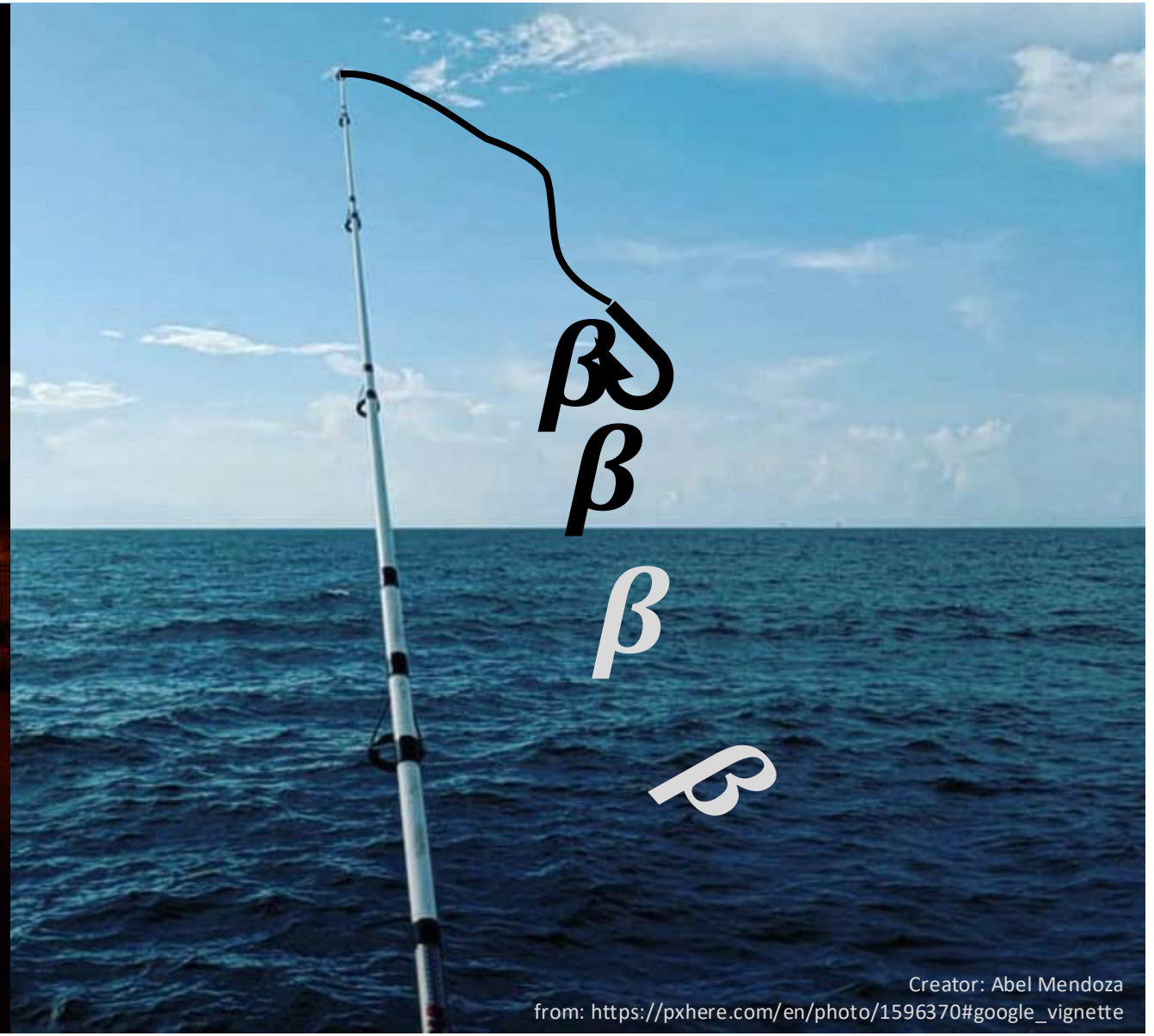
- Include all input variables that might affect the outcome.
- Several predictors can be combined into a sum or average score.
- For inputs with large effects, consider their interactions.
- On the exclusion of variables:
  - If predictor is not statistically significant and has the expected sign, leave it in the model.
  - If a predictor is not statistically significant and does not have the expected sign, consider removing it.
  - If a predictor is statistically significant and does not have the expected sign, think hard and consider collecting more data.
  - If a predictor is statistically significant and has the expected sign, keep it in.

Gelman & Hill 2007

Automating multiple linear regression



Creator: Michael Wyszomierski
from: https://www.flickr.com/photos/wysz/10215555

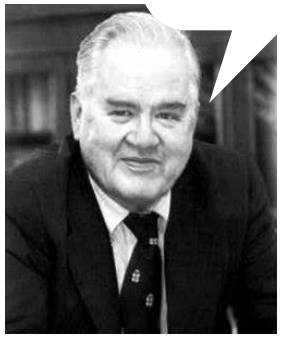


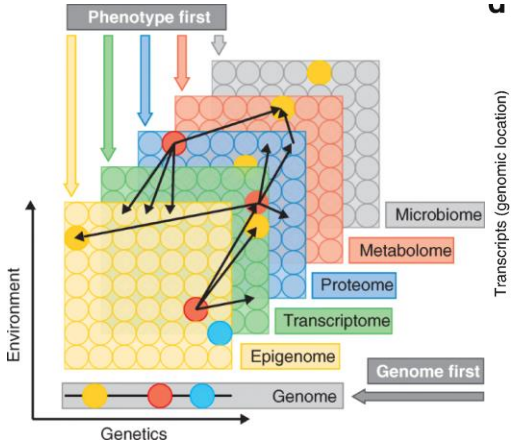
Creator: Abel Mendoza
from: https://pxhere.com/en/photo/1596370#google_vignette

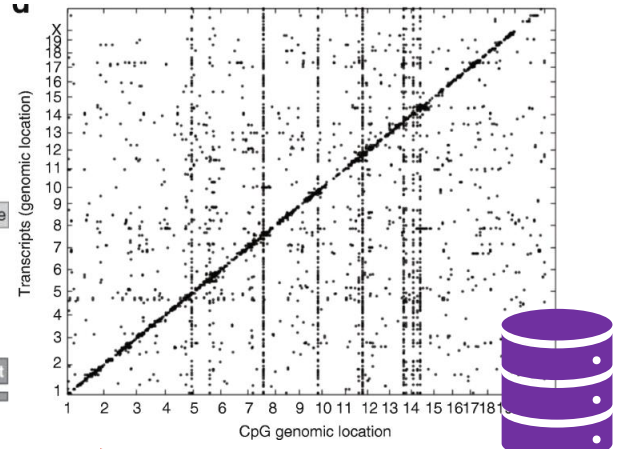
Automating multiple linear regression

Exploratory data analysis [EDA] can never be the whole story, but nothing else can serve as the foundation stone – as the first step.

[EDA] is detective work – numerical detective work -- or counting detective work – or graphical detective work.
John Tukey 1977







from: Hasin-Brumshstein *et al.* 2017

EDA & Data mining

- 1960s data mining
- 1977 – Tukey, Exploratory Data Analysis
- 1990s data discovery

Prediction

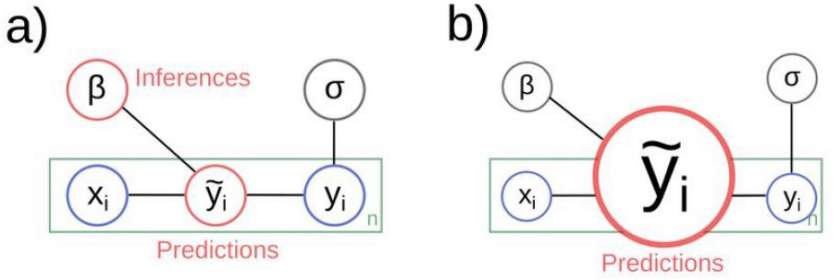
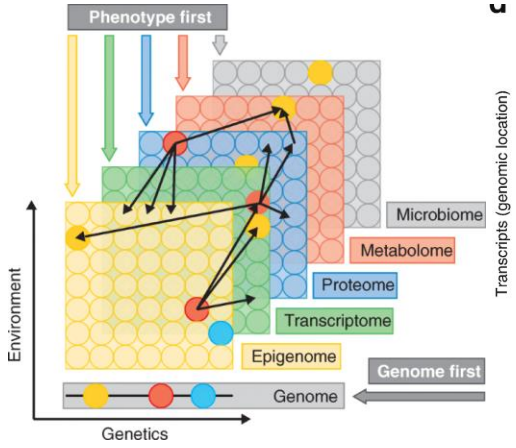
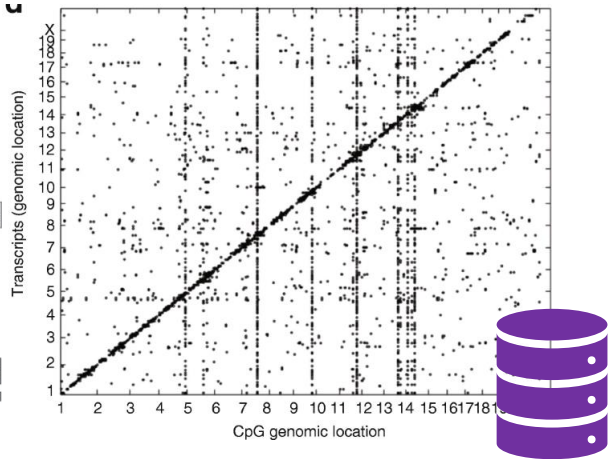


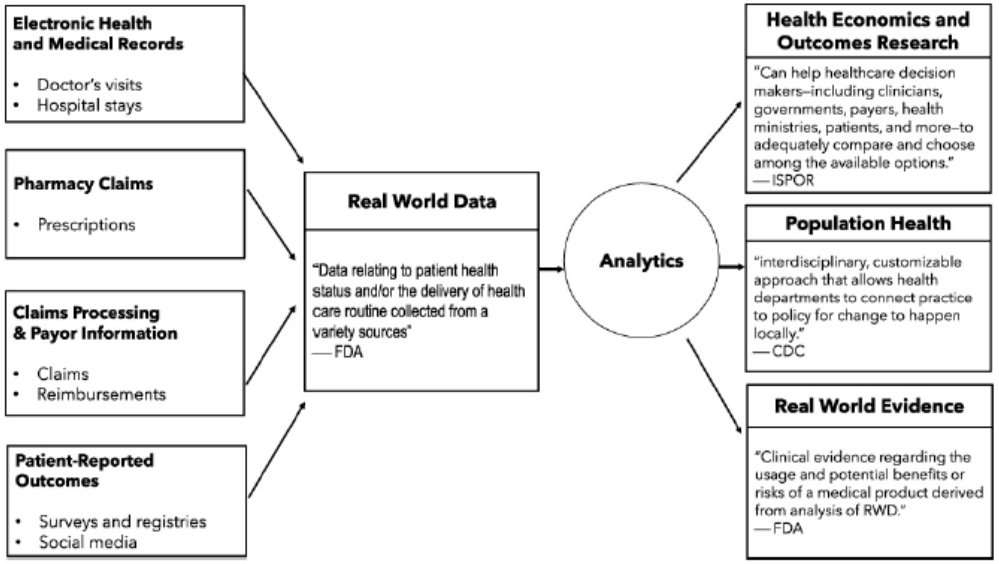
Figure 2. Illustration of perspectives on the sample model of Figure 1: a) a balanced perspective equally valuing inference and prediction and b) a prediction-oriented viewpoint.

Sanders 2019

Regression modelling and real-world data







from: Hasin-Brumshstein *et al*. 2017; Zou *et al*. 2020

Multiple linear regression model:

$$y = \beta_0 + \beta_1 \cdot x_1 \ldots + \beta_p \cdot x_p + \epsilon$$

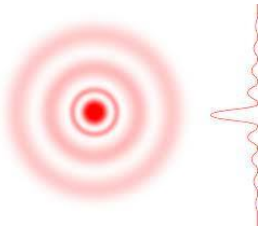
Dataset with ***p*** features and ***i*** samples:

$$p \cdot i, \qquad p > i$$

Our data will not support estimating all of these effects (degrees of freedom).
Often high collinearity present.
Even large samples can become tricky if we solely rely on statistical metrics to guide inclusion of effectors (overfitting).

The proteomic dataset

In a proteomic dataset of 20 quantified biomarkers, what are the best predictors of disease activity?



disease activity

,

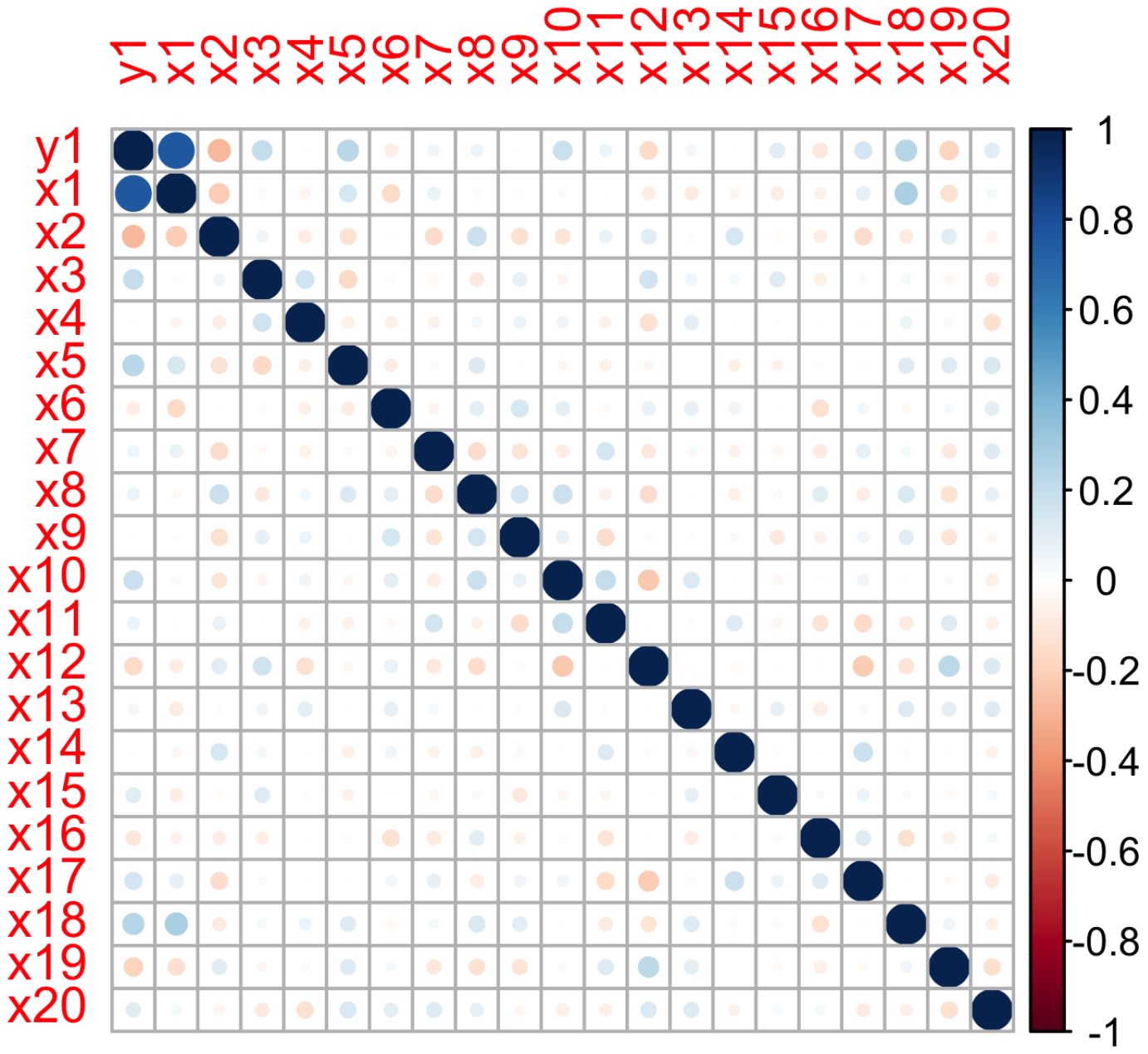


N=100 samples

X



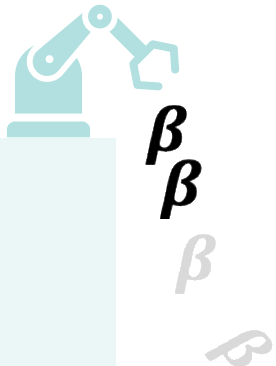
N=20 protein activities




```r
# Correlation plot
library(corrplot)

cor_dat = cor(omicsdata)
corrplot(cor_dat)
```

Automated variable, or feature, selection



Subset selection

- Best subset selection, 2^p models to evaluate.
- Forward and and backward stepwise selection, approximately p^2 models.

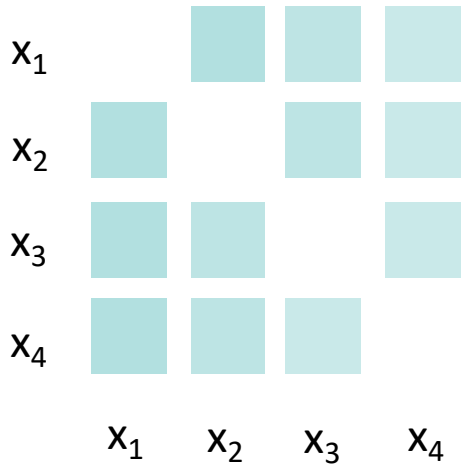
Shrinkage/regularisation methods penalising for the size of coefficients

- **Ridge** shrinks Beta coefficients close to zero.
- **Lasso** creates sparse models (Beta coefficients = 0).
- **Elastic net** a mix between Ridge and Lasso.

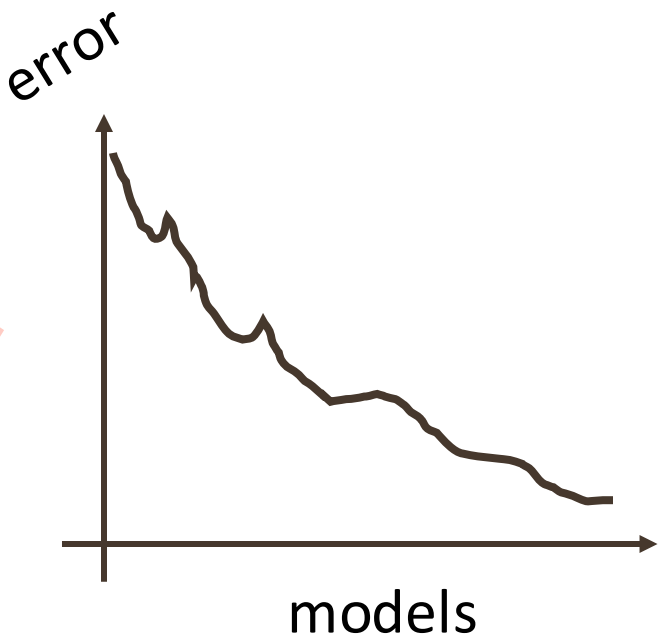
Dimension reduction methods

- Principal component regression

Best subset selection



finite set of X with n elements
then: $\mathfrak{S}_n$ or 2^p models



Model evaluation according to criterion

Bayesian Information Criterion

Adjusted R^2

Mallows' Cp

k-fold cross validation

Criteria

$$SS_{error} = \sum_{i=1}^{n} \left(y_i - f(x; \hat{\theta})\right)^2$$

$$AIC = N \cdot \ln\left(\frac{SS_{error}}{N}\right) + 2 \cdot K$$

$$BIC = N \cdot \ln\left(\frac{SS_{error}}{N}\right) + K \cdot \ln(N)$$

$$\bar{R}^2 = 1 - (1 - R^2) \cdot \frac{N - 1}{N - K - 1}$$

$$C_p = \frac{1}{N}(SS_{error} + 2p\hat{\sigma}^2)$$

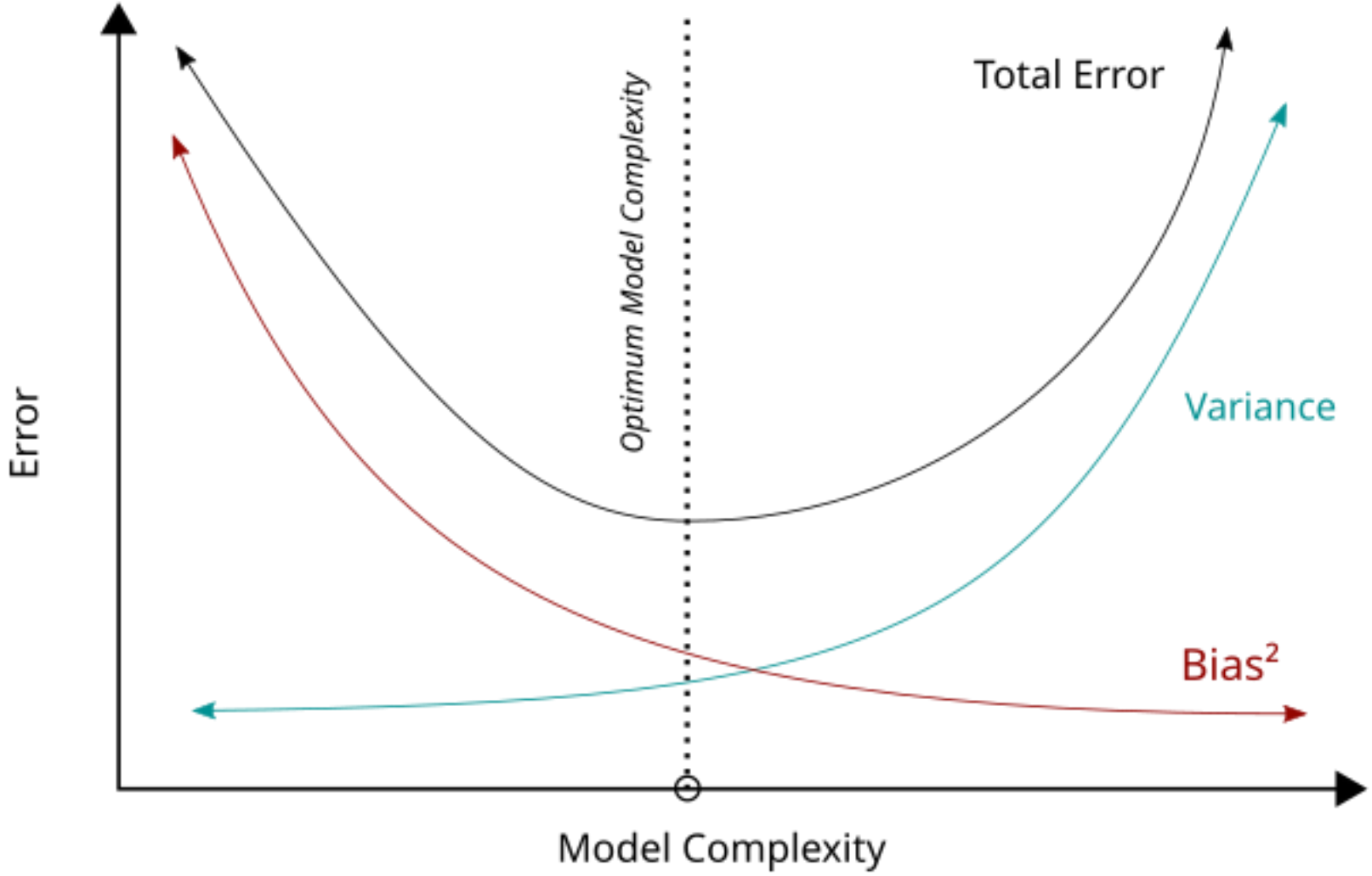
- **Aikake Information Criterion (AIC)** Trade-off between goodness-of-fit of the model and model complexity.

  N: number of observations,

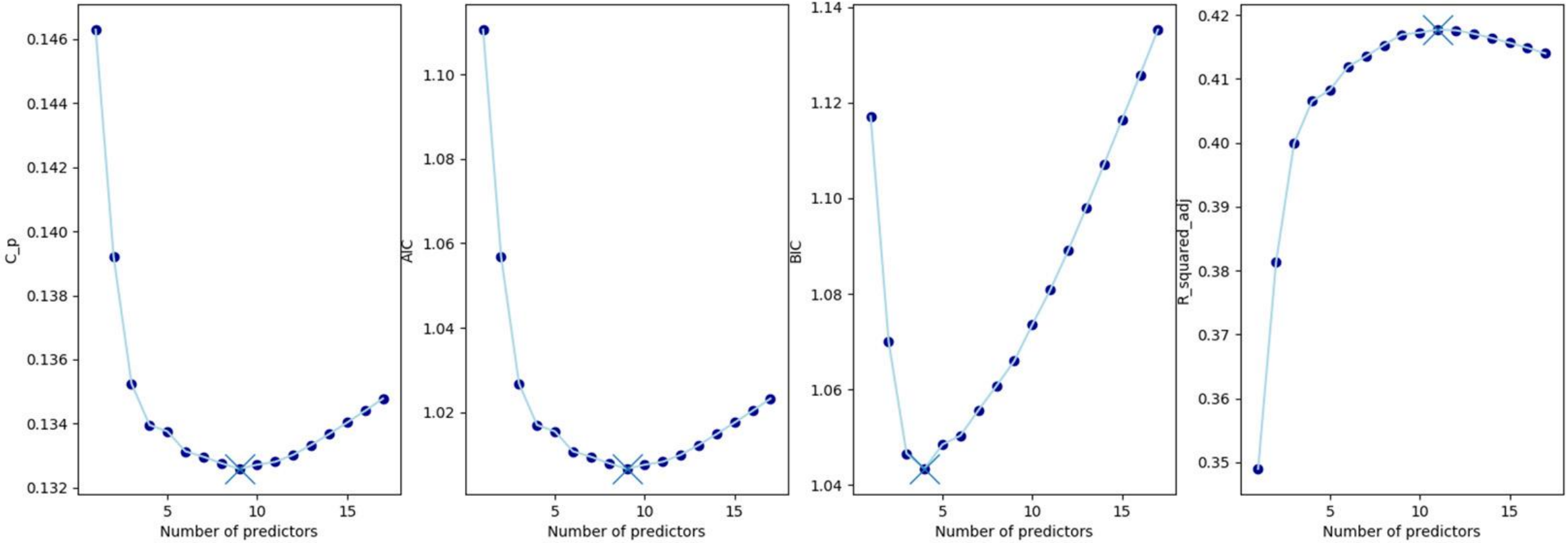
  SS_{error}: sum of squares error,

  K: N parameters + 1
- The lower the value of AIC the better is the goodness-of-fit relative to model complexity.
- **Bayesian Information Criterion (BIC)** penalises N parameters to a higher extent than the AIC.
- **Adjusted R^2** indicates how well the model explains the data, also considering number of parameters in the model.
- **Mallows' Cp** Compares precision and bias between the full model and subsets (the smaller the better).

Bias-variance tradeoff



Criteria



Woo & Kim 2022

Best subset selection in R

R

```
library(leaps)# package for best subset

# run best subset on the dataset
M1 <- regsubsets(omicsdata$y1~., data = omicsdata, nvmax = 20)
summary(M1)

# model selection, e.g.: adjusted R^2, Mallows' CP, BIC
ResM1 <- summary(M1)
data.frame(
  Adj.R2 = which.max(ResM1$adjr2),
  CP = which.min(ResM1$cp),
  BIC = which.min(ResM1$bic)
)
```

Best subset selection in R

```
1 subsets of each size up to 20 Selection Algorithm: exhaustive
x1 x2 x3 x4 x5 x6 x7 x8 x9 x10 x11 x12 x13 x14 x15 x16 x17 x18 x19 x20
1 ( 1 ) "*" " " "
```

k-fold cross validation

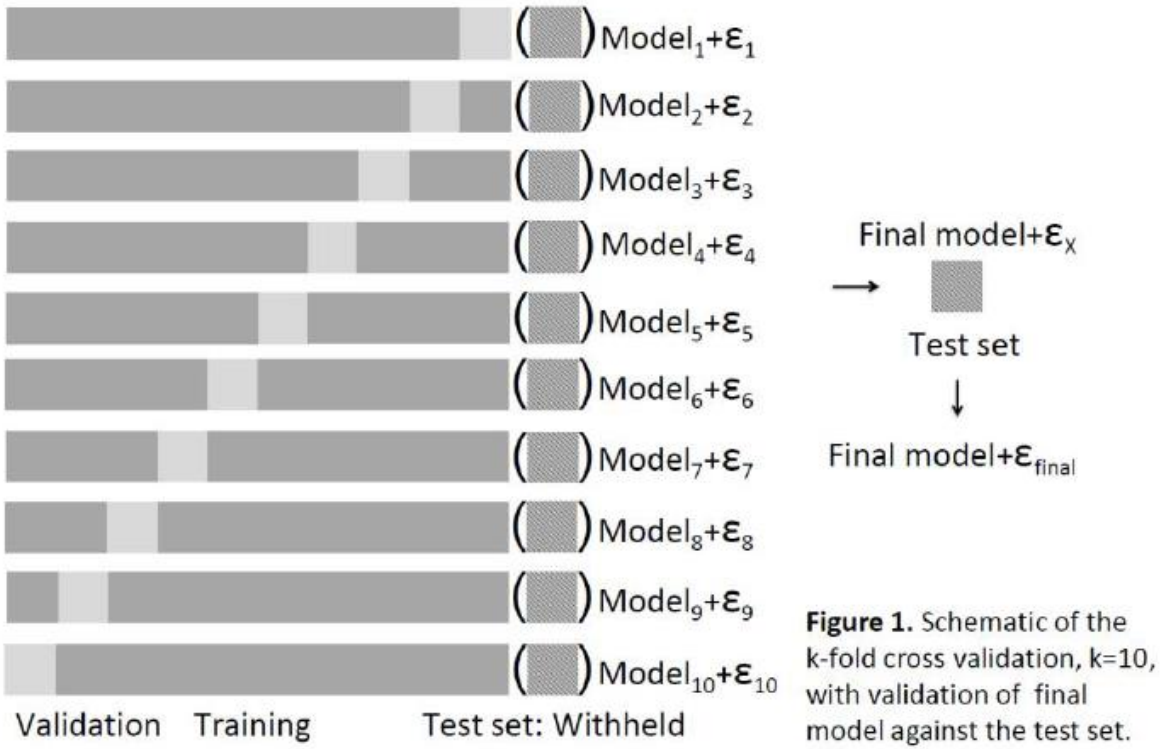
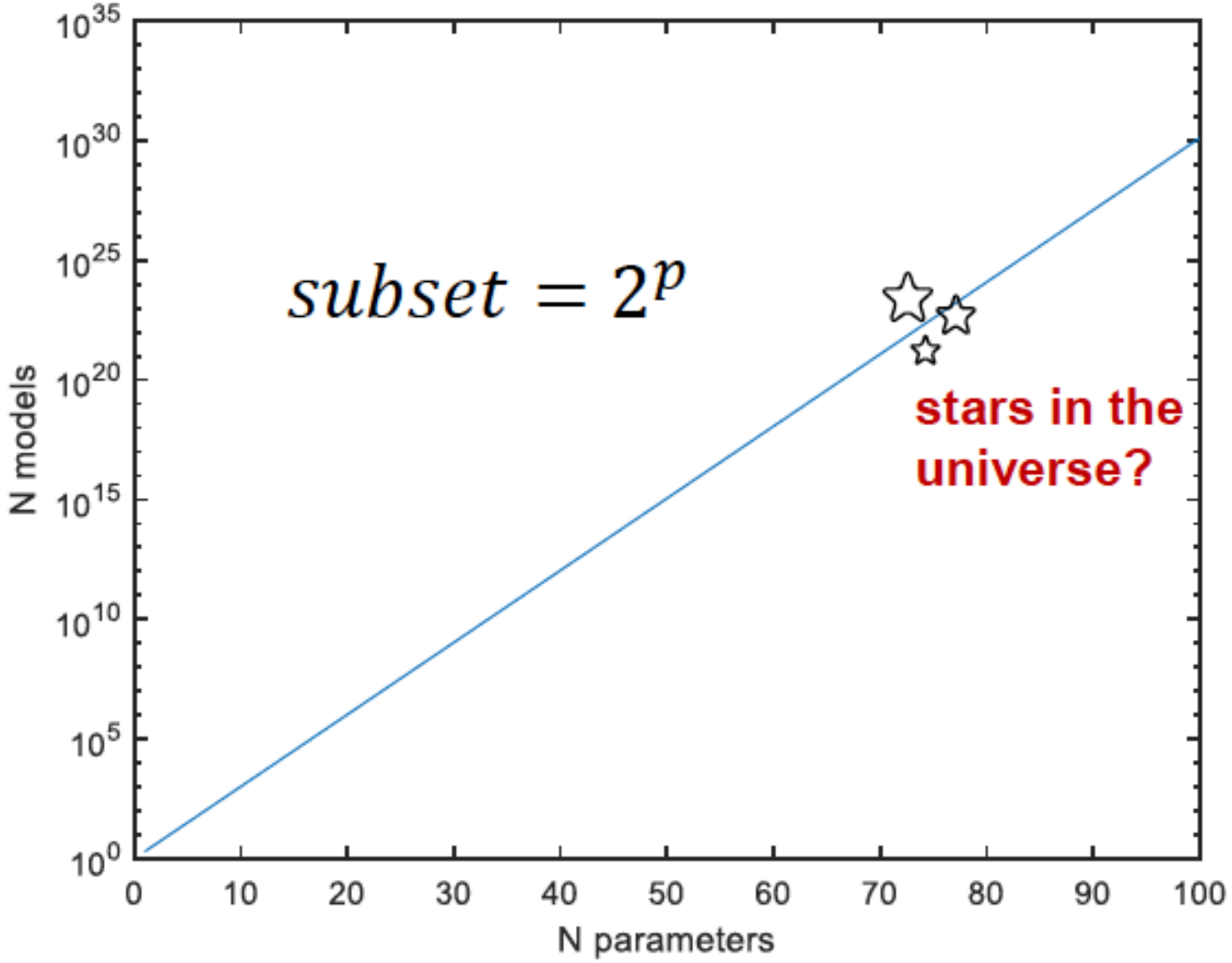


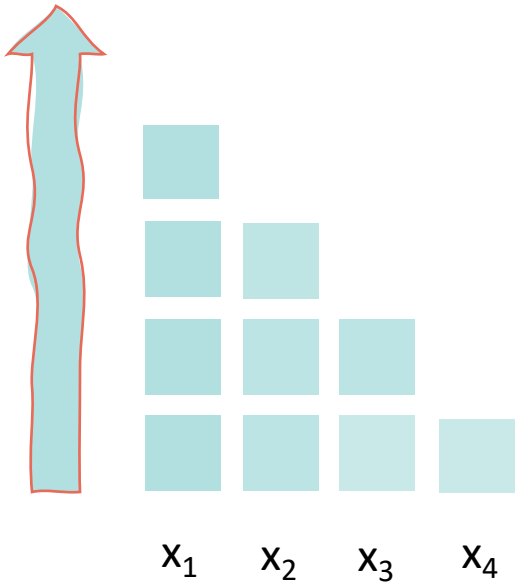
Figure 1. Schematic of the k-fold cross validation, k=10, with validation of final model against the test set.

Darwich *et al*. 2015

```
(Intercept)x1          x7          x15
54.64,      2.32,      -2.31,      2.22
```


$subset = 2^p$
stars in the universe?
N models
N parameters
10^{35}
10^{30}
10^{25}
10^{20}
10^{15}
10^{10}
10^{5}
10^{0}
0
10
20
30
40
50
60
70
80
90
100

Backward stepwise selection

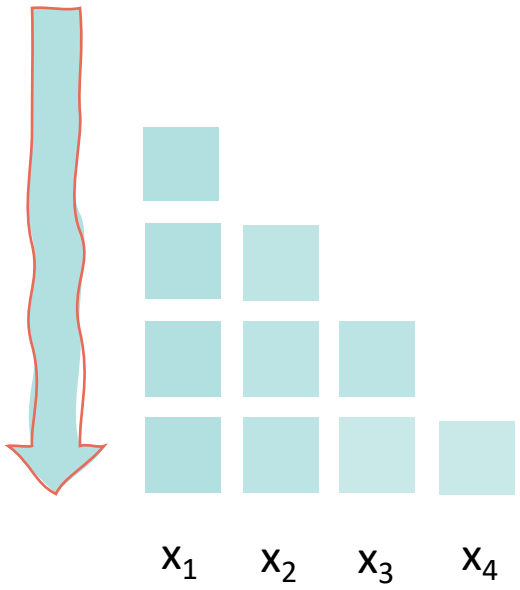


- We start with the full least squares model, M_p, that contains all ***p*** independent variables.
- Then we work our way backwards by choosing the best model for ***p - 1***, and so on, until we reach model M_0, the null-model, with only an intercept:

$$1 + p \cdot (p + 1)/2$$

- We can then compare across models and select the best one according to our criteria.
- We encounter a problem if we have more variables than observations, ***p>i***, then we cannot estimate a unique solution.

Forward stepwise selection



- We start from the null model, M_0.
- We add predictors to the model, one parameter at-a-time.
- In each step, we chose the model that gives the best improvement in performance.
- Models are nested, statistically comparable.
- Computational advantage over best subset.
- Not guaranteed to find the “best” model compared to best subset. Because it might not be nested.
- This does not mean that the stepwise model will perform any worse in prediction exercises.

Forward stepwise selection

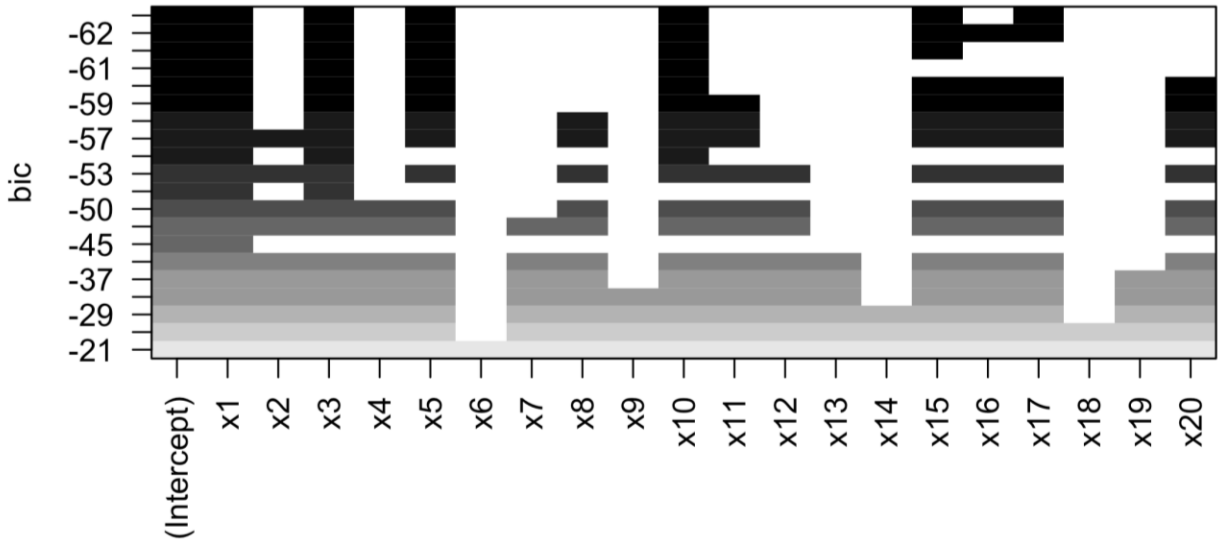
```
# load libraries
library(leaps)

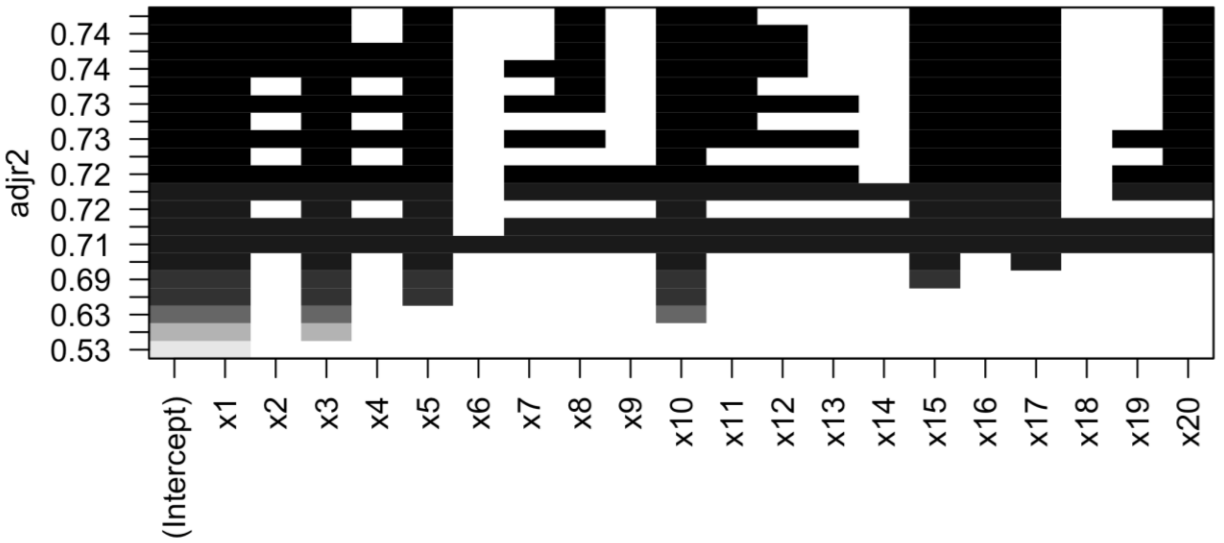
# load dataset
omicsdata1 <- read.table(file="omicsdata.csv", header=TRUE, sep = ",")
attach(omicsdata1)

set.seed(123456) # set random seed
# sample 70 indices from 100
train <- sample(seq(100),70,replace = FALSE)

fwdfit <- regsubsets(DV~.,data=omicsdata1[train,],nvmax = 20, method = "forward")
```





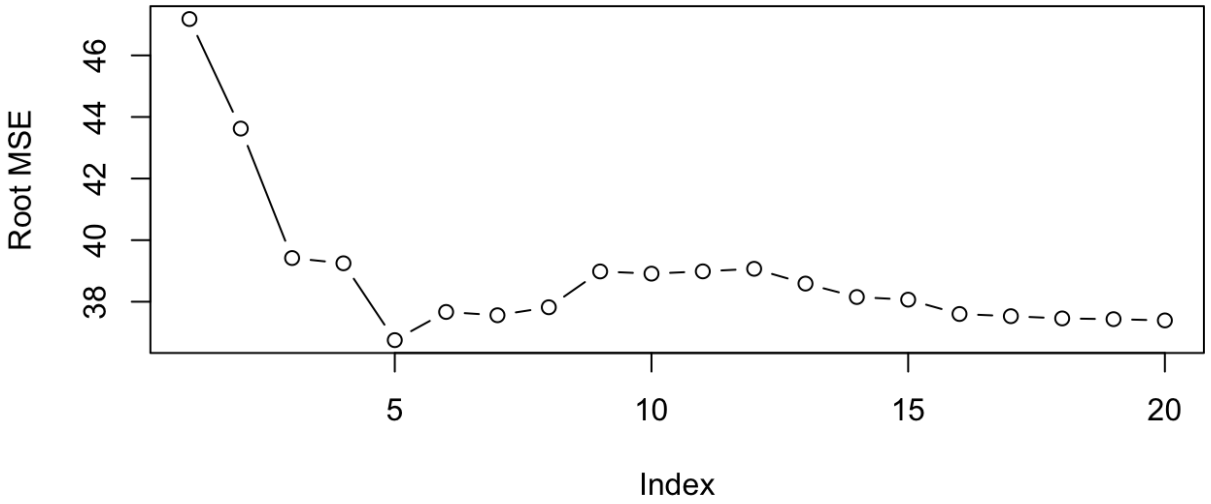


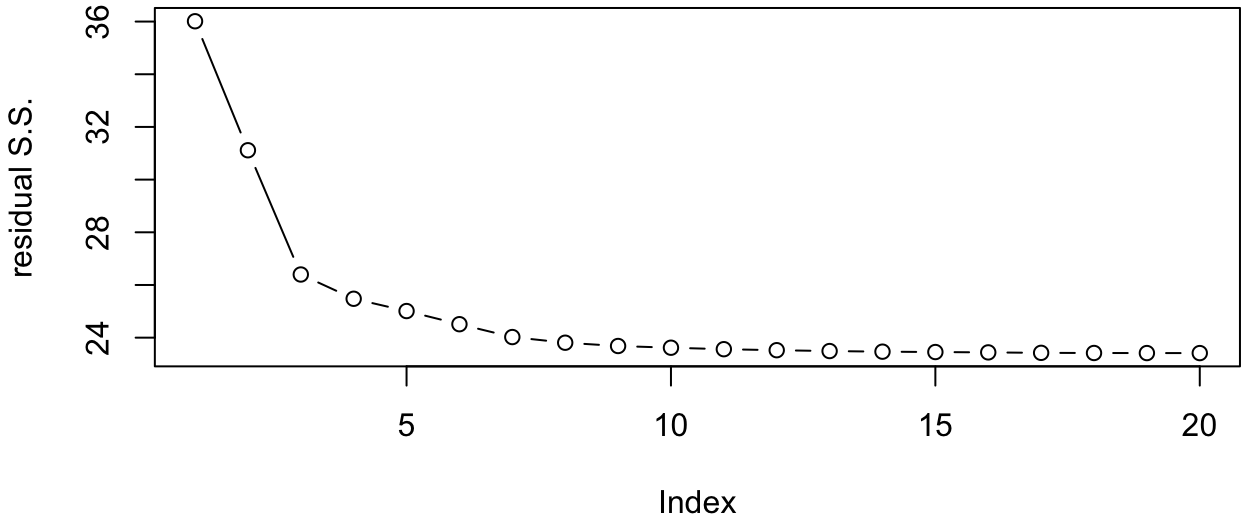

```
# plot model selection criteria
plot(fwdfit) # Bayesian Information Criteria
plot(fwdfit,scale="Cp") # Mallows' Cp / "AIC"
plot(fwdfit,scale="adjr2") # Adjusted R^2

# find "best" models
fwdfitsum <- summary(fwdfit)
data.frame(
  Adj.R2 = which.max(fwdfitsum$adjr2),
  CP = which.min(fwdfitsum$cp),
  BIC = which.min(fwdfitsum$bic)
)
```

Adj.R2 <int>	CP <int>	BIC <int>
11	11	6

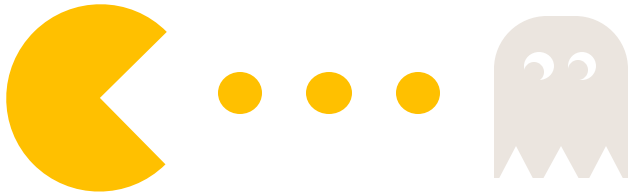
Validation






```
validerr <- rep(NA,20)
xtest <- model.matrix(DV~.,data=omicsdata1[-train,])
for(i in 1:20){
  coefi <- coef(fwdfit,id=i)
  pred <- xtest[,names(coefi)]%*%coefi
  validerr[i] <- mean((omicsdata1$DV[-train]-pred)^2)
}
plot(sqrt(validerr),ylab="Root MSE",type="b")
plot(sqrt(fwdfit2$rss[-1]/70),ylab="residual S.S.",type="b")
```

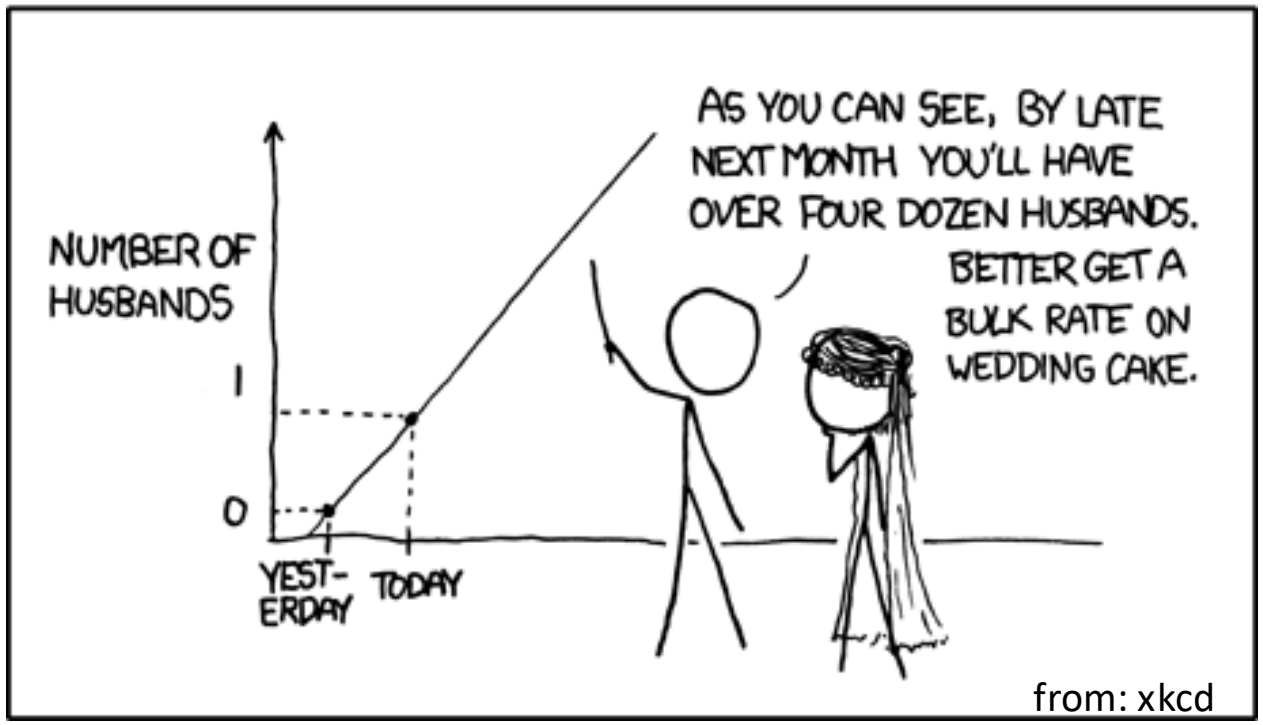
What is the problem with stepwise regression?



$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j \cdot x_{i,j} \right)^2$$

- At any given step the model is fitted using unconstrained least squares.
- *p*-values are too low due to multiple comparisons across models, this is difficult to correct for in this approach.
- Biased, generally too large, parameter estimates.
- Collinearity creates problems.
- A greedy approach.

ref: Harrell 2001; Gelman 2014

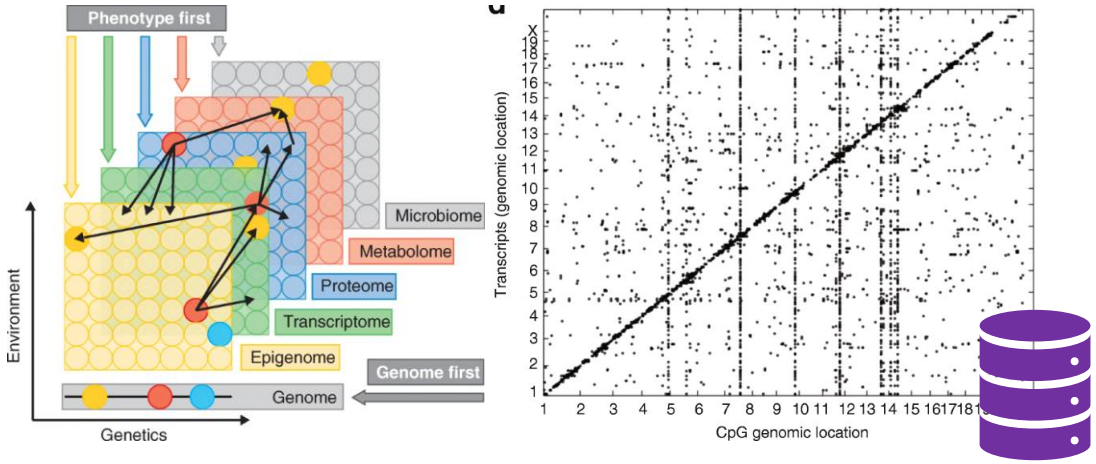


If our aim is to make predictions, how can we find a reliable, sparse model?

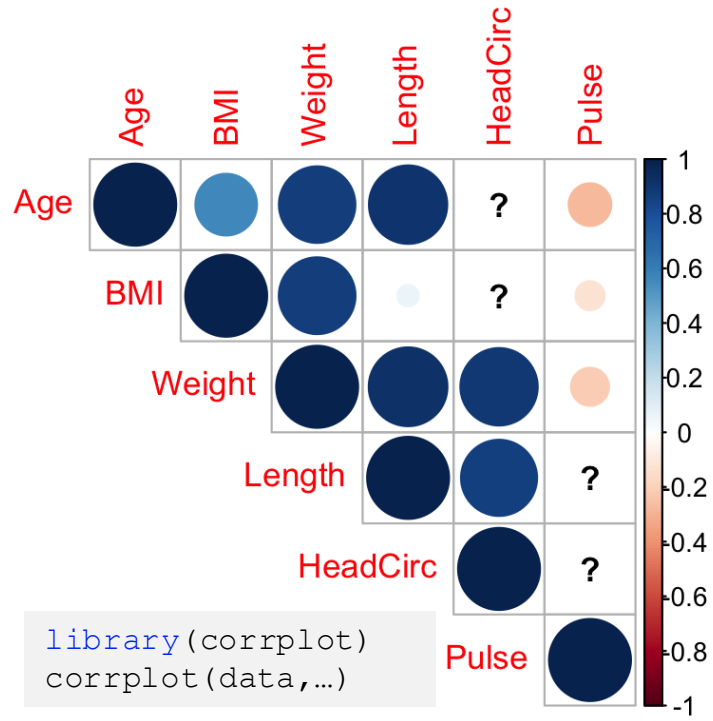
L5-2 Multiple linear regression

Ridge regression

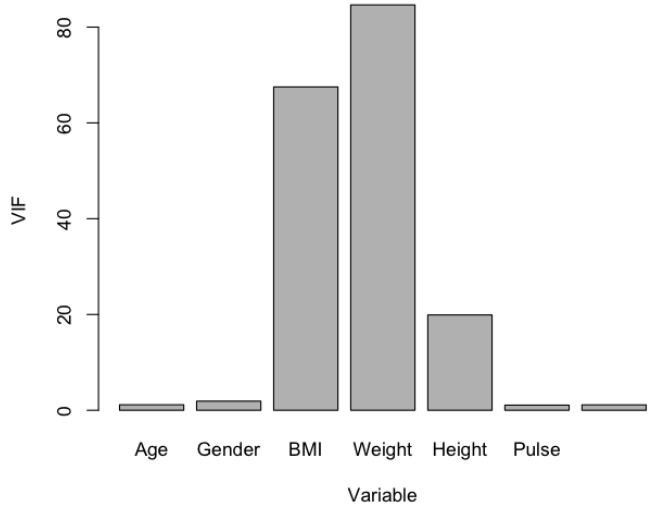
On the issue of collinearity



- Multicollinearity is a prominent issue in real-world data
- In earlier videos we looked at ways of detecting collinearity
- Data correlation matrices, r close to one indicates collinearity
- Model Variance Inflation Factor (VIF) where VIF over 10 suggests an issue

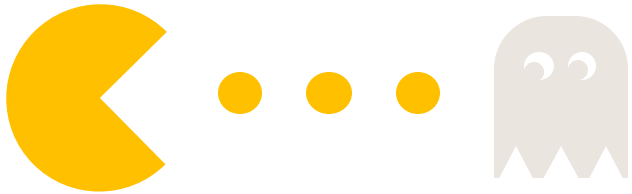



```
library(corrplot)
corrplot(data,…)
```




```
library(car), vif(model)
```

What is the problem with stepwise regression?



$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j \cdot x_{i,j} \right)^2$$

- At any given step the model is fitted using unconstrained least squares.
- *p*-values are too low due to multiple comparisons across models, this is difficult to correct for in this approach.
- Biased, generally too large, parameter estimates.
- Collinearity problems.
- Very greedy approach.

ref: Harrell 2001; Gelman 2014

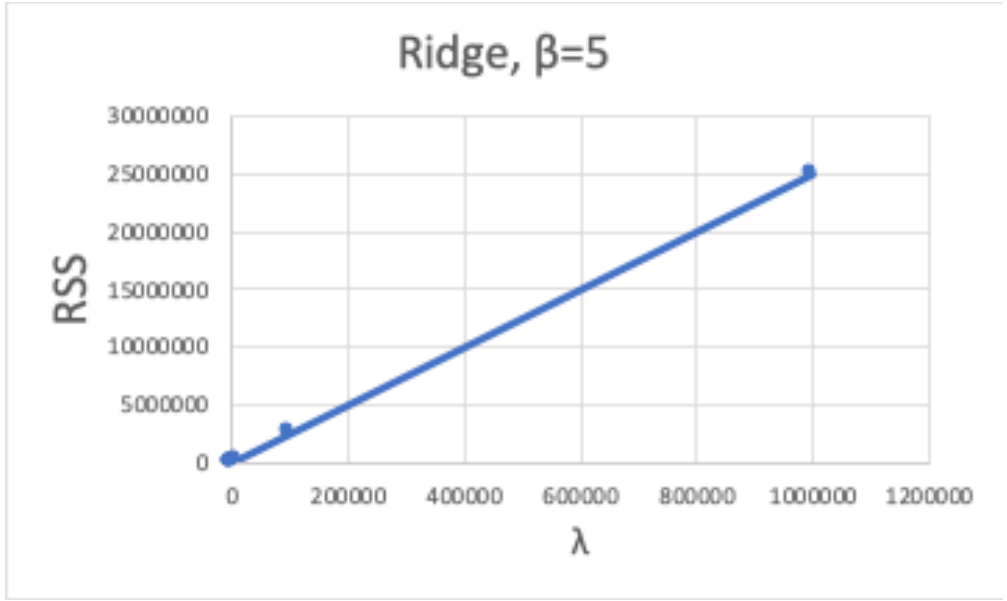
Shrinkage methods: Ridge regression

- Fit model using all $\boldsymbol{p}$ predictors.
- The model regularises and shrinks the estimates of the coefficients towards zero.
- $\boldsymbol{\lambda}$ is adjusted to investigate the impact of level of regularisation.
- We can then determine the value of $\boldsymbol{\lambda}$ through k-fold cross validation.

ref: Hoerl and Kennard 1970; Hastie, *et al*. 2017

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j\cdot x_{ij}\right)^2+\lambda\cdot\sum_{j=1}^{p}\beta_j^2=RSS+\lambda\cdot\sum_{j=1}^{p}\beta_j^2$$

Shrinkage methods: Ridge regression



$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j\cdot x_{ij}\right)^2+\lambda\cdot\sum_{j=1}^{p}\beta_j^2=RSS+\lambda\cdot\sum_{j=1}^{p}\beta_j^2$$

Centering and/or standardising data

```
Age <- Age – mean(age)
```

Centering Linear transformations, such as the above, do not affect residual errors or R^2. They can simplify the interpretation of regressions, differences from the mean.

```
Age <- (Age – mean(age))/(2*sd(Age))
```

Standardising Scaling of predictors allows comparisons across coefficients on approximately the same scale.

Darwich *et al*. 2015

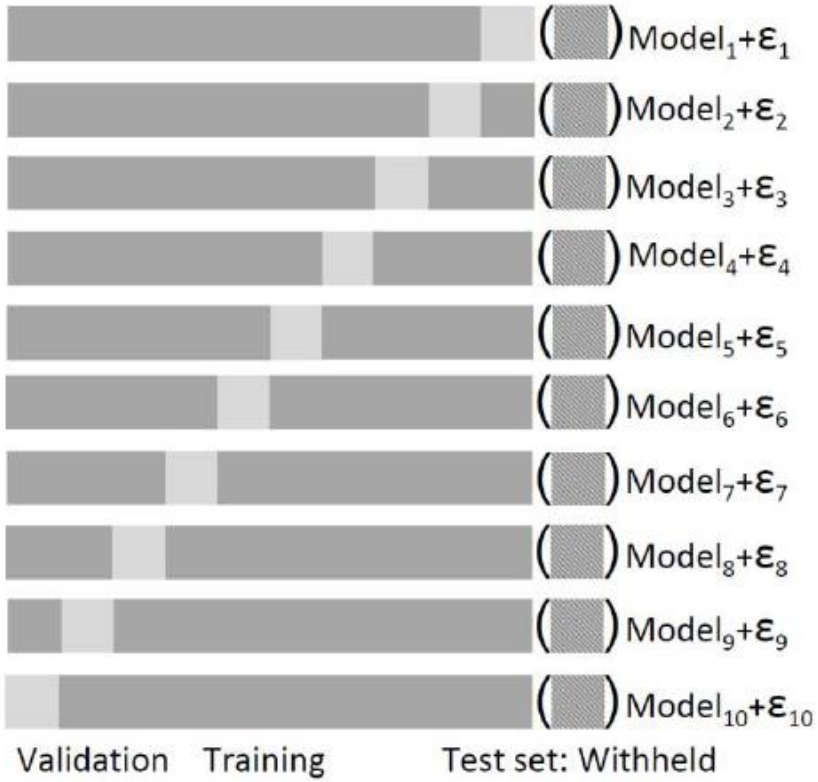
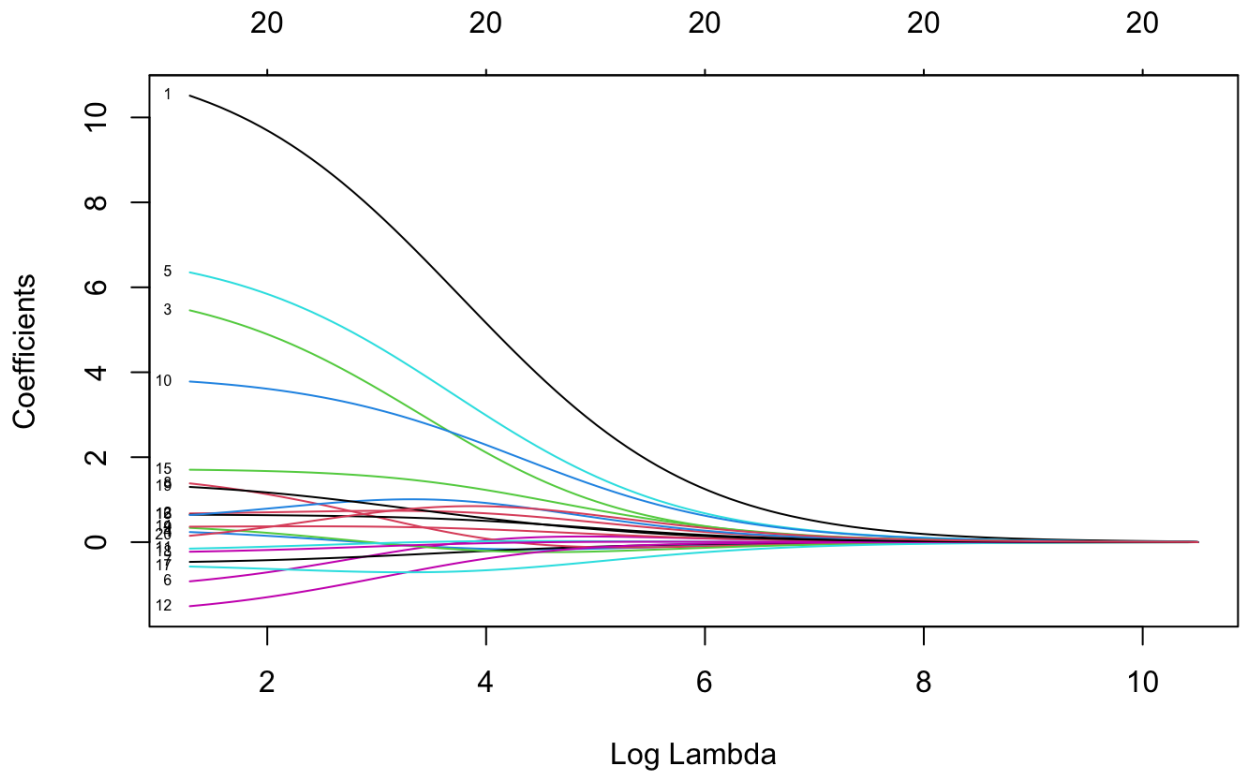
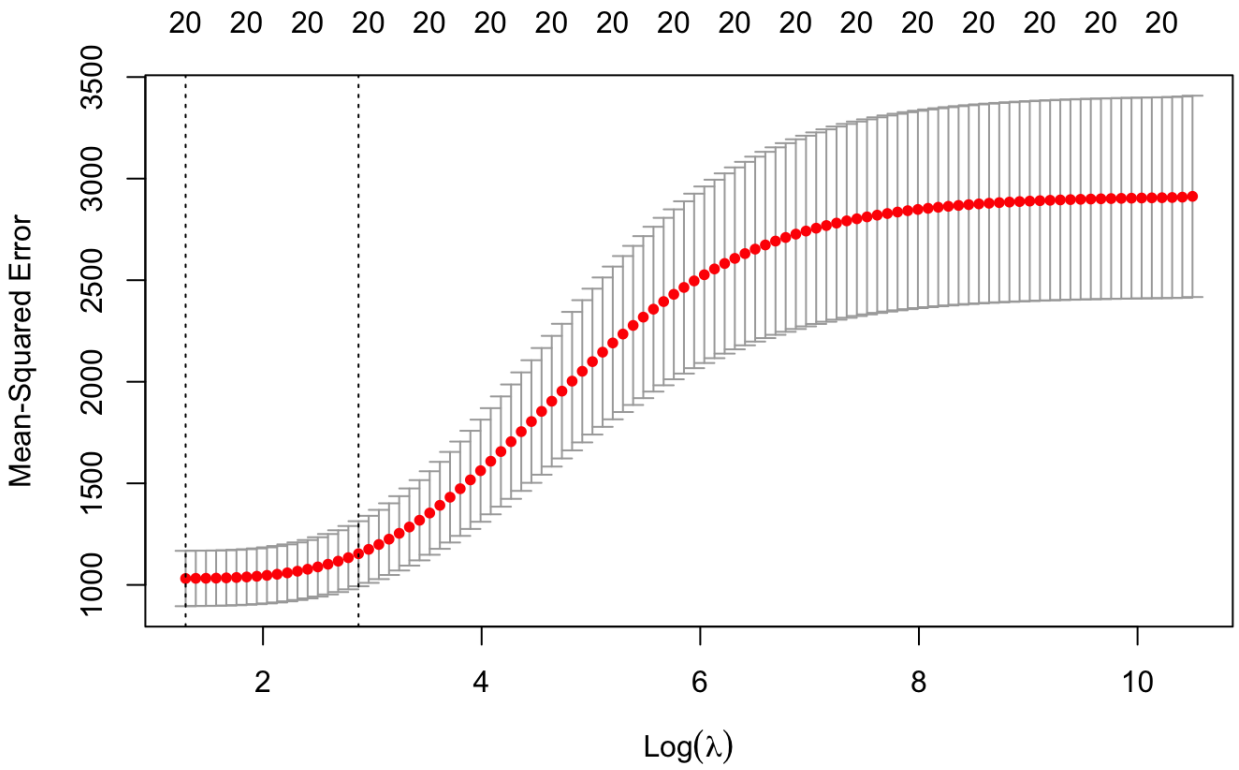


Figure 1. Schematic of the k-fold cross validation, k=10, with validation of final model against the test set.

```
library(glmnet)
x=model.matrix(DV~.-1,data=omicsdata1)
y=omicsdata1$DV

ridgefit1 <- glmnet(x,y,alpha=0,standardize=TRUE)
plot(ridgefit1,xvar="lambda",label=TRUE)
ridgecv1 <- cv.glmnet(x,y,alpha=0) # 10-fold cross
validation
plot(ridgecv1)
coef(ridgecv1)
```


20 20 20 20 20
Coefficients
0 2 4 6 8 10
1
5
3
10
15
6
12
2 4 6 8 10
Log Lambda


20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
Mean-Squared Error
1000 1500 2000 2500 3000 3500
2 4 6 8 10
Log(λ)

R

```
(Intercept) 58.289016035
x1           8.050374115
x2           0.735809408
x3           3.812530705
x4          -0.003014166
x5           4.806854913
x6          -0.350326914
x7          -0.343907472
x8           0.666184363
x9           0.015735504
x10          3.207655489
x11         -0.035429550
x12         -0.909576554
x13          0.604415439
x14          0.367604488
x15          1.570401236
x16          0.974628192
x17         -0.697938018
x18         -0.112313709
x19          0.926840307
x20          0.661608490
```

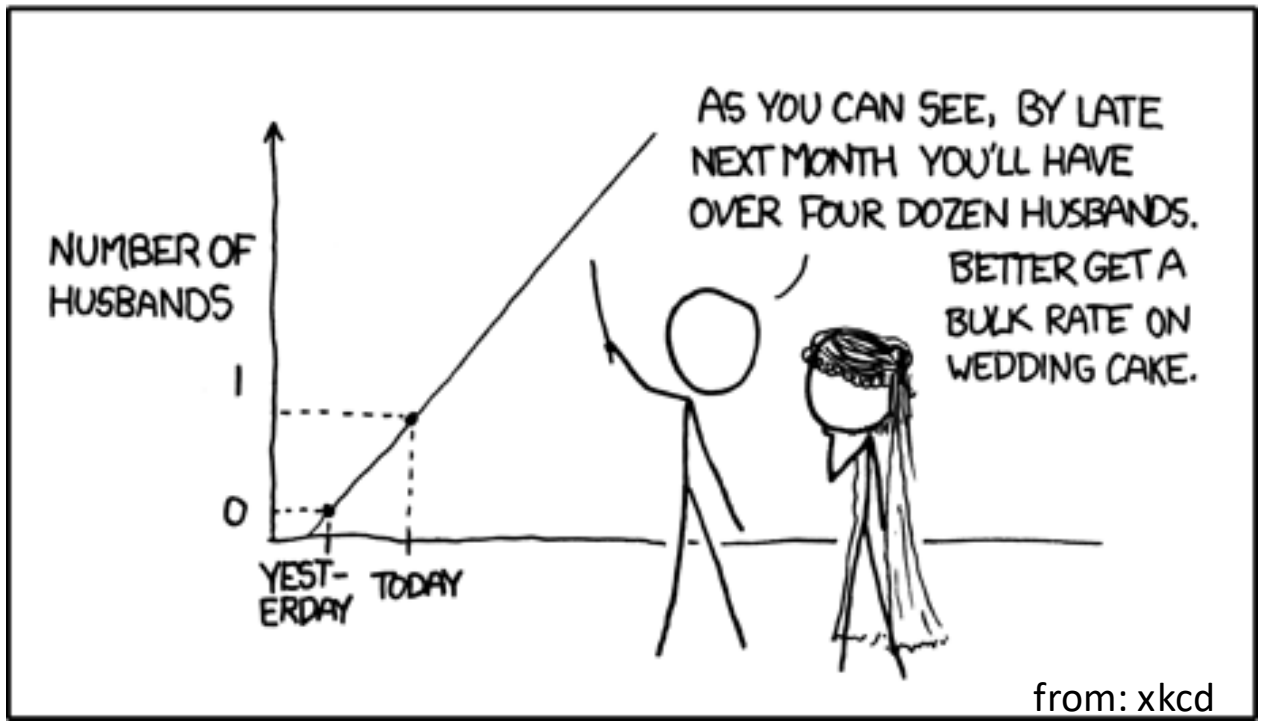


L5-3 Multiple linear regression

Lasso regression

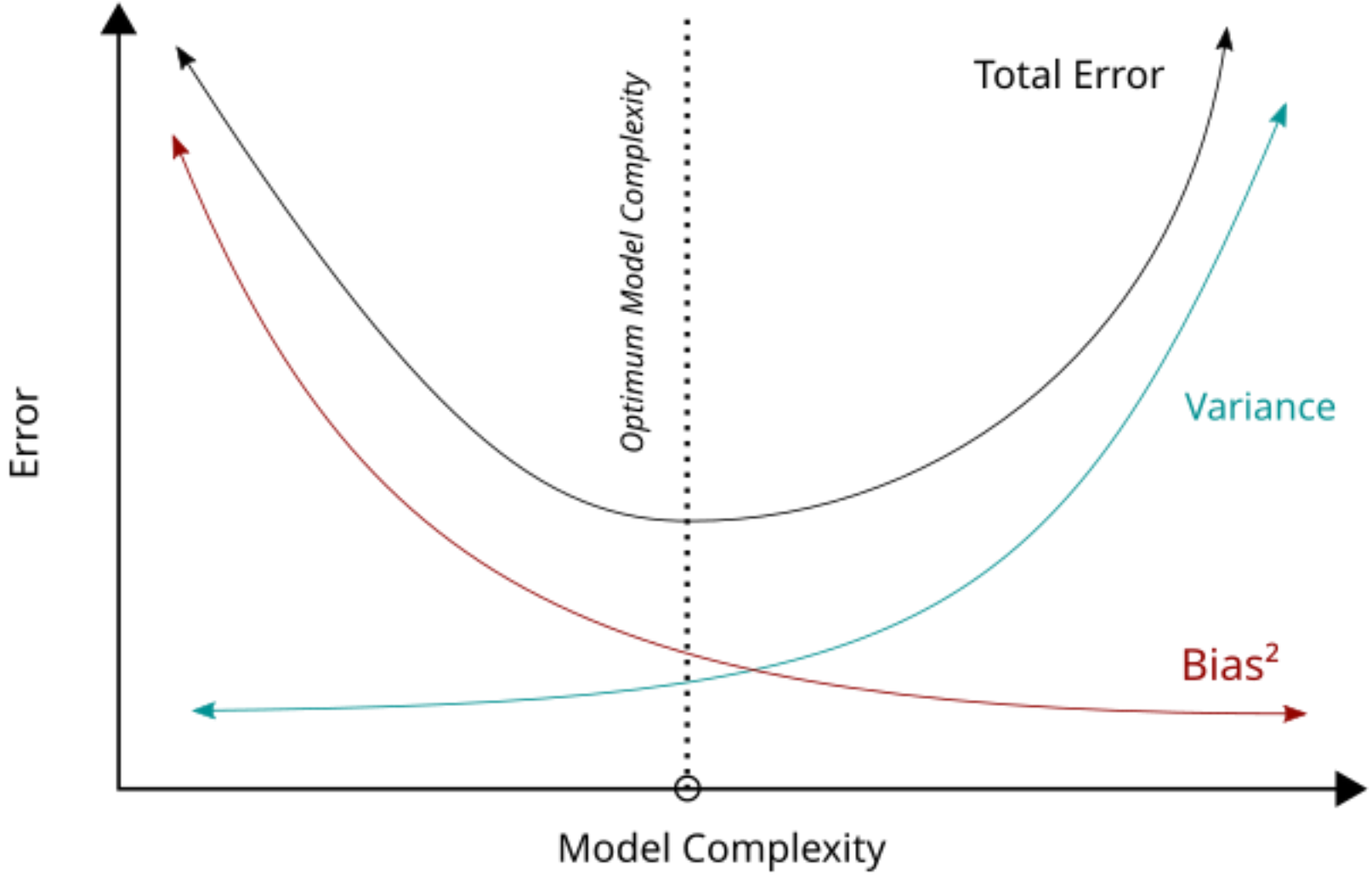
R

```
(Intercept) 132.8253476
x1            8.2900718
x2           -1.7212333
x3            2.9692469
x4           -0.1177422
x5            2.4051377
x6           -0.1845882
x7           -0.2689658
x8            0.6806413
x9           -0.3189365
x10           1.8338230
x11           0.9874573
x12          -0.9118595
x13           0.5788732
x14           0.1584552
x15           1.4961599
x16          -0.9396511
x17           1.0167898
x18           0.8383614
x19          -1.3010770
x20           1.1330688
```



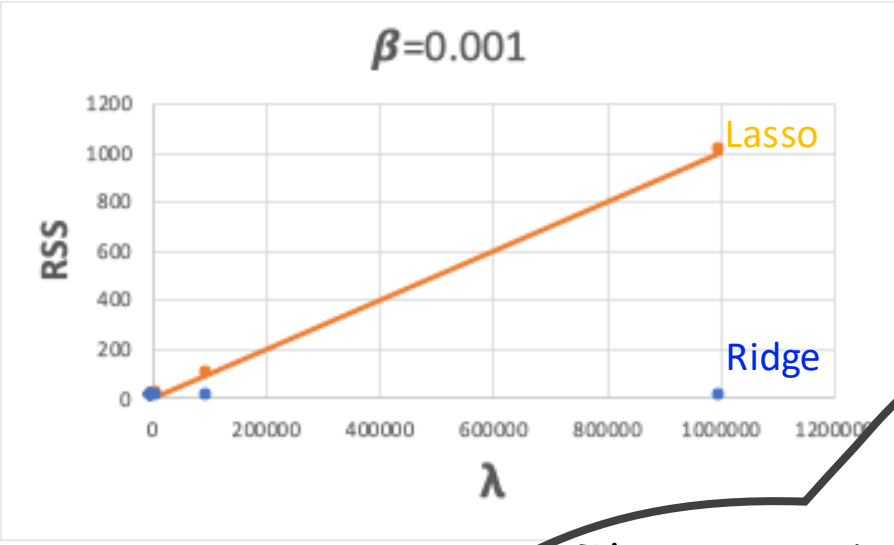
If our aim is to make predictions, how can we find a reliable, sparse model?

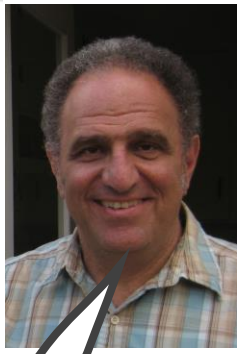
Bias-variance tradeoff



Shrinkage methods: Lasso regression

- Ridge regression retains all coefficients ($\beta_j>0$) even as they approach zero.
- The Lasso will shrink coefficients to zero and therefore carrious out variable selection (through $\beta_j=0$).
- Creates sparse, predictive regression models.
- Lasso will not remedy collinearity.





The Least Absolute Shrinkage and Selection Operator, Lasso.

Tibshirani 1996; Tibshirani 1997

$$\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j\cdot x_{ij}\right)^2+\lambda\cdot\sum_{j=1}^{p}|\beta_j|=RSS+\lambda\cdot\sum_{j=1}^{p}|\beta_j|$$

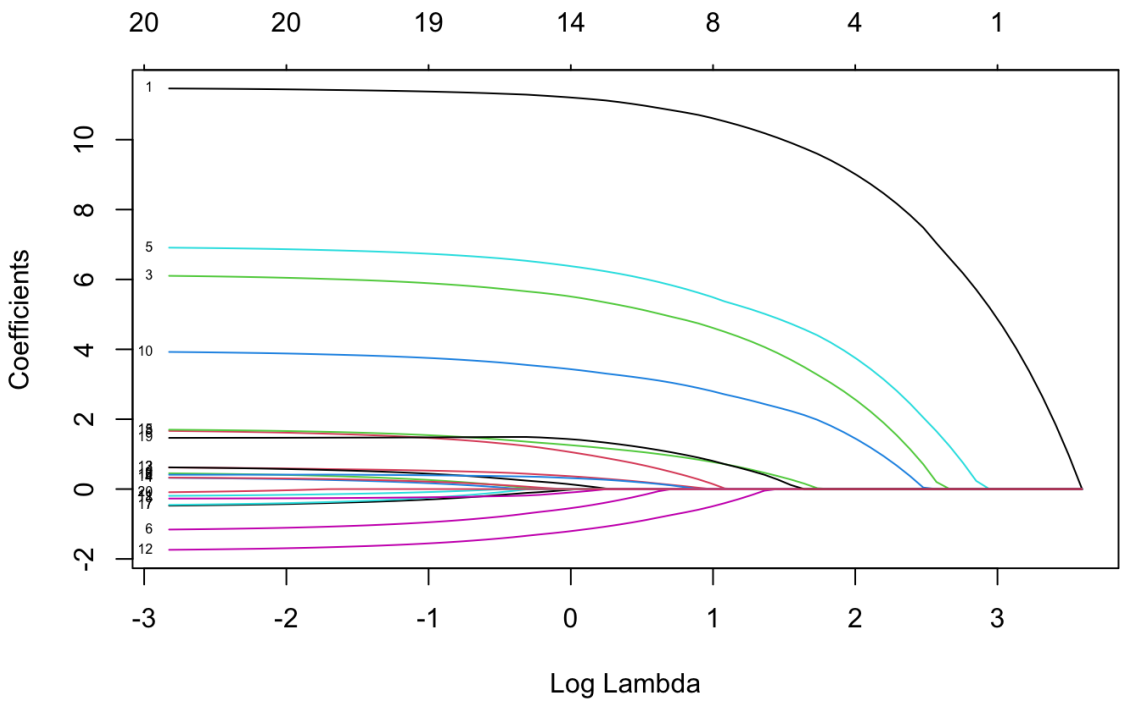
Ridge

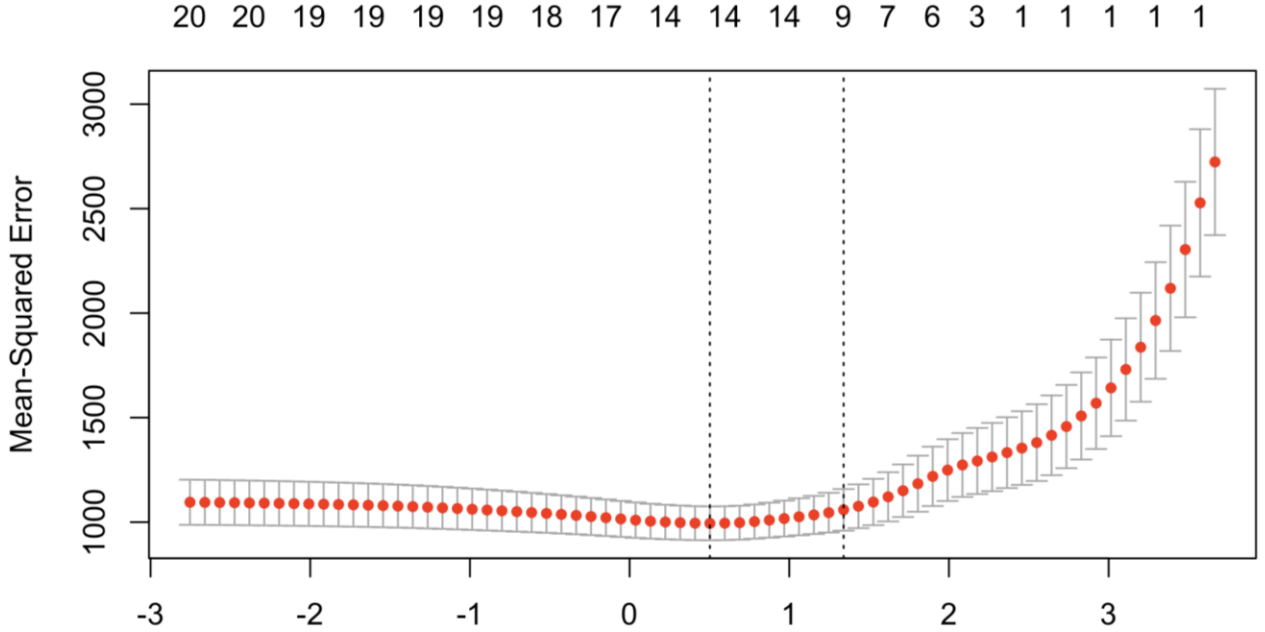
$$-2X^T(y-X\beta)+2\lambda\beta=0$$

Lasso

$$\frac{d}{d\beta}(\lambda|\beta|)=\begin{cases}\lambda & \beta>0\\ -\lambda & \beta<0\\ [-\lambda,\lambda] & \beta=0\end{cases}$$

Lasso






```
lassofit1 <- glmnet(x,y, standardize=TRUE)
plot(lassofit1,xvar="lambda",label=TRUE)
lassocv1 <- cv.glmnet(x,y)
plot(lassocv1)
coef(lassocv1)
```

Ridge

R

```
(Intercept) 132.8253476
x1            8.2900718
x2           -1.7212333
x3            2.9692469
x4           -0.1177422
x5            2.4051377
x6           -0.1845882
x7           -0.2689658
x8            0.6806413
x9           -0.3189365
x10           1.8338230
x11           0.9874573
x12          -0.9118595
x13           0.5788732
x14           0.1584552
x15           1.4961599
x16          -0.9396511
x17           1.0167898
x18           0.8383614
x19          -1.3010770
x20           1.1330688
```

Lasso

R

```
(Intercept) 114.9783540
x1           11.5533249
x2           -0.5968255
x3            2.8440778
x4            .
x5            1.7657989
x6            .
x7            .
x8            .
x9            .
x10           1.7914453
x11           .
x12          -0.5056282
x13           .
x14           .
x15           1.2965136
x16           .
x17           .
x18           .
x19          -0.1377573
x20           0.2879115
```

L5-4 Multiple linear regression

Elastic Net

R

(Intercept)	58.289016035
x1	8.050374115
x2	0.735809408
x3	3.812530705
x4	-0.003014166
x5	4.806854913
x6	-0.350326914
x7	-0.343907472
x8	0.666184363
x9	0.015735504
x10	3.207655489
x11	-0.035429550
x12	-0.909576554
x13	0.604415439
x14	0.367604488
x15	1.570401236
x16	0.974628192
x17	-0.697938018
x18	-0.112313709
x19	0.926840307
x20	0.661608490

R

(Intercept)	86.2689059
x1	9.9099687
x2	.
x3	3.6990587
x4	.
x5	4.7321304
x6	.
x7	.
x8	.
x9	.
x10	2.2334492
x11	.
x12	.
x13	.
x14	.
x15	0.2714038
x16	.
x17	.
x18	.
x19	0.1140254
x20	.

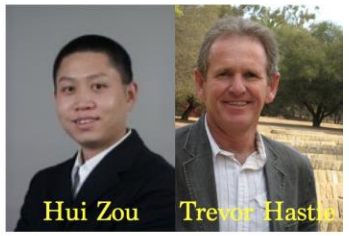
Elastic net the best of two worlds?

Regularization and variable selection via the elastic net

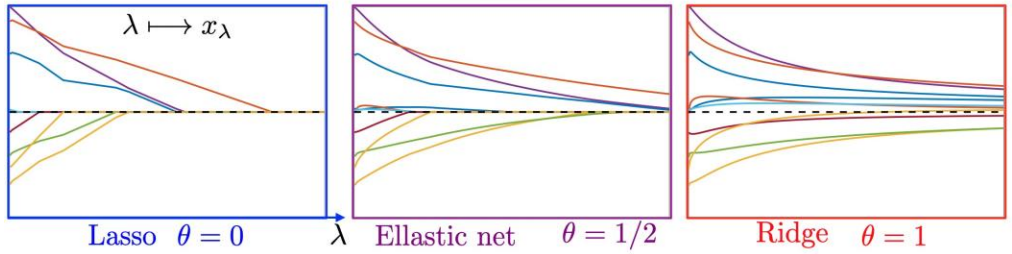
Hui Zou and Trevor Hastie
Stanford University, USA

[Received December 2003. Final revision September 2004]

Summary. We propose the elastic net, a new regularization and variable selection method. Real world data and a simulation study show that the elastic net often outperforms the lasso, while enjoying a similar sparsity of representation. In addition, the elastic net encourages a grouping effect, where strongly correlated predictors tend to be in or out of the model together. The elastic net is particularly useful when the number of predictors (p) is much bigger than the number of observations (n). By contrast, the lasso is not a very satisfactory variable selection method in the $p \gg n$ case. An algorithm called LARS-EN is proposed for computing elastic net regularization paths efficiently, much like algorithm LARS does for the lasso.



$$\text{Elastic net:} \quad x_\lambda \in \operatorname*{argmin}_x \frac{1}{2\lambda}\|Ax-y\|^2 + (1-\theta)\|x\|_1 + \frac{\theta}{2}\|x\|_2^2$$



- Ridge (alpha=0) is good at dealing with multicolinearity by penalising for larger coefficients and shrinking these → β^2.
- Lasso (alpha=1) is good at creating sparse models (variable selection) where the penality term, $|\beta|$, leads to β-coefficients equal to zero.
- Elastic net is a mix between the two. Can deal with collinearity and create sparse models.

$$RSS + \lambda \cdot \sum_{j=1}^{p} \left(\frac{1}{2} \cdot (1-\alpha) \cdot \beta^2 + \alpha \cdot |\beta| \right)$$

- Vary and finding best alpha value through k-fold cross validation:

Zou and Hastie 2005

Example at:
James et al. 2023. Statistical Learning with Applications in R. Second Edition.

Tutorial available at:
Engebretsen S. & Bohlin J. Statistical predictions with glmnet. Clinical Epigenetics 11, 123 (2019).

```
cv_table <- sample(1:10,size=length(y), replace(true))
elasticnetfit1 <- cv.glmnet(x,y, cv_table=cv_table, alpha=0.0)
elasticnetfit1 <- cv.glmnet(x,y, cv_table=cv_table, alpha=0.1)
..
```

Elastic Net

```
library(caret)
library(glmnet)

set.seed(123456)

# train model 5-fold cross validation (James et al. 2023)
cv_5 = trainControl(method = "cv", number = 5)

hit_elnet_int = train(
  y1 ~ ., data = omicsdata,
  method = "glmnet",
  trControl = cv_5,
  tuneLength = 10
)

# get model with best tuning parameters
get_best_result = function(caret_fit) {
  best = which(rownames(caret_fit$results) == rownames(caret_fit$bestTune))
  best_result = caret_fit$results[best, ]
  rownames(best_result) = NULL
  best_result
}

get_best_result(hit_elnet_int)

# run CV glmnet with alpha = 1, i.e. Lasso
fit_cv = cv.glmnet(x, y, alpha = 1)
sqrt(fit_cv$cvm[fit_cv$lambda == fit_cv$lambda.min])
# Based on minimum mean square error in the cross validation

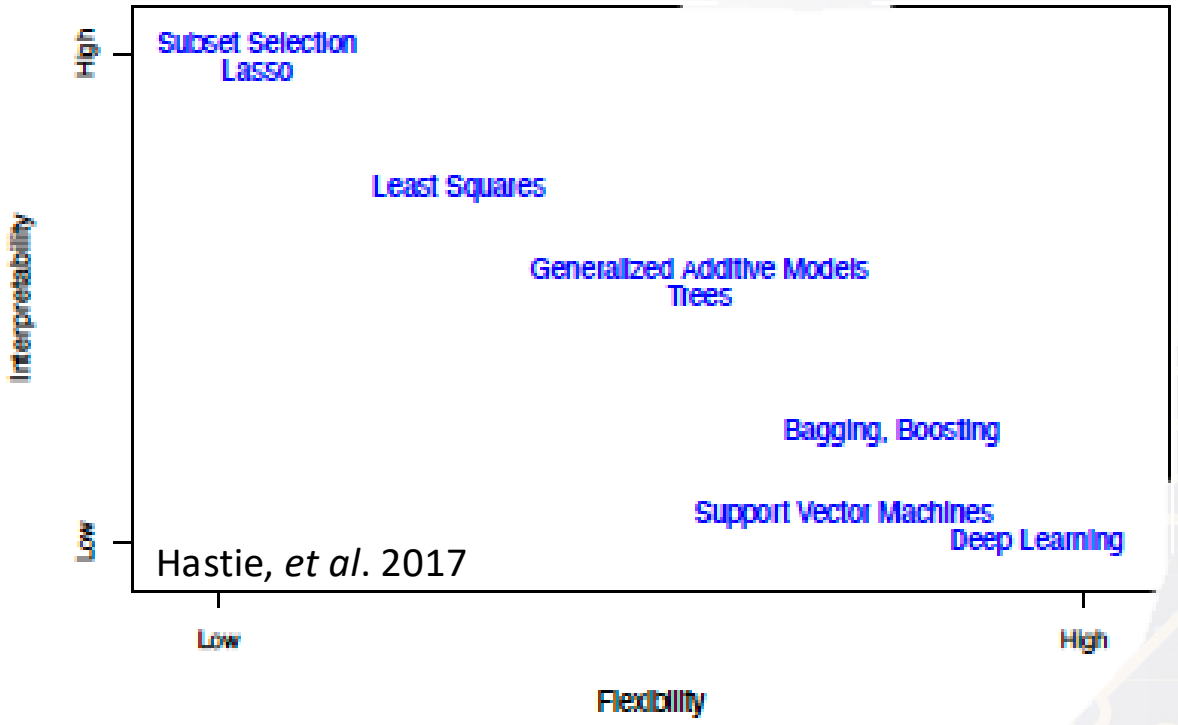
sum(coef(fit_cv) != 0)
```

Example at:
[James et al. 2023. Statistical Learning with Applications in R. Second Edition.]

Lasso

alpha <dbl>	lambda <dbl>	RMSE <dbl>	Rsquared <dbl>	MAE <dbl>	RMSESD <dbl>	RsquaredSD <dbl>	MAESD <dbl>
1	2.745561	32.78892	0.580747	26.63568	1.612314	0.1706177	2.69537

Interpretability vs. Flexibility



Hastie, *et al*. 2017

FIGURE 2.7. *A representation of the tradeoff between flexibility and interpretability, using different statistical learning methods. In general, as the flexibility of a method increases, its interpretability decreases.*

Warning



MIT Technology Review Topics

Tech policy / AI Ethics

The UK exam debacle reminds us that algorithms can't fix broken systems

The problem began when the exam regulator lost sight of the ultimate goal—and pushed for standardization above all else.

by **Karen Hao** August 20, 2020

https://www.technologyreview.com/2020/08/20/1007502/uk-exam-algorithm-cant-fix-broken-system/



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NEWS · 24 OCTOBER 2019 · UPDATE 26 OCTOBER 2019

Millions of black people affected by racial bias in health-care algorithms

Study reveals rampant racism in decision-making software used by US hospitals — and highlights ways to correct it.

https://www.nature.com/articles/d41586-019-03228-6

Could this be a remedy?

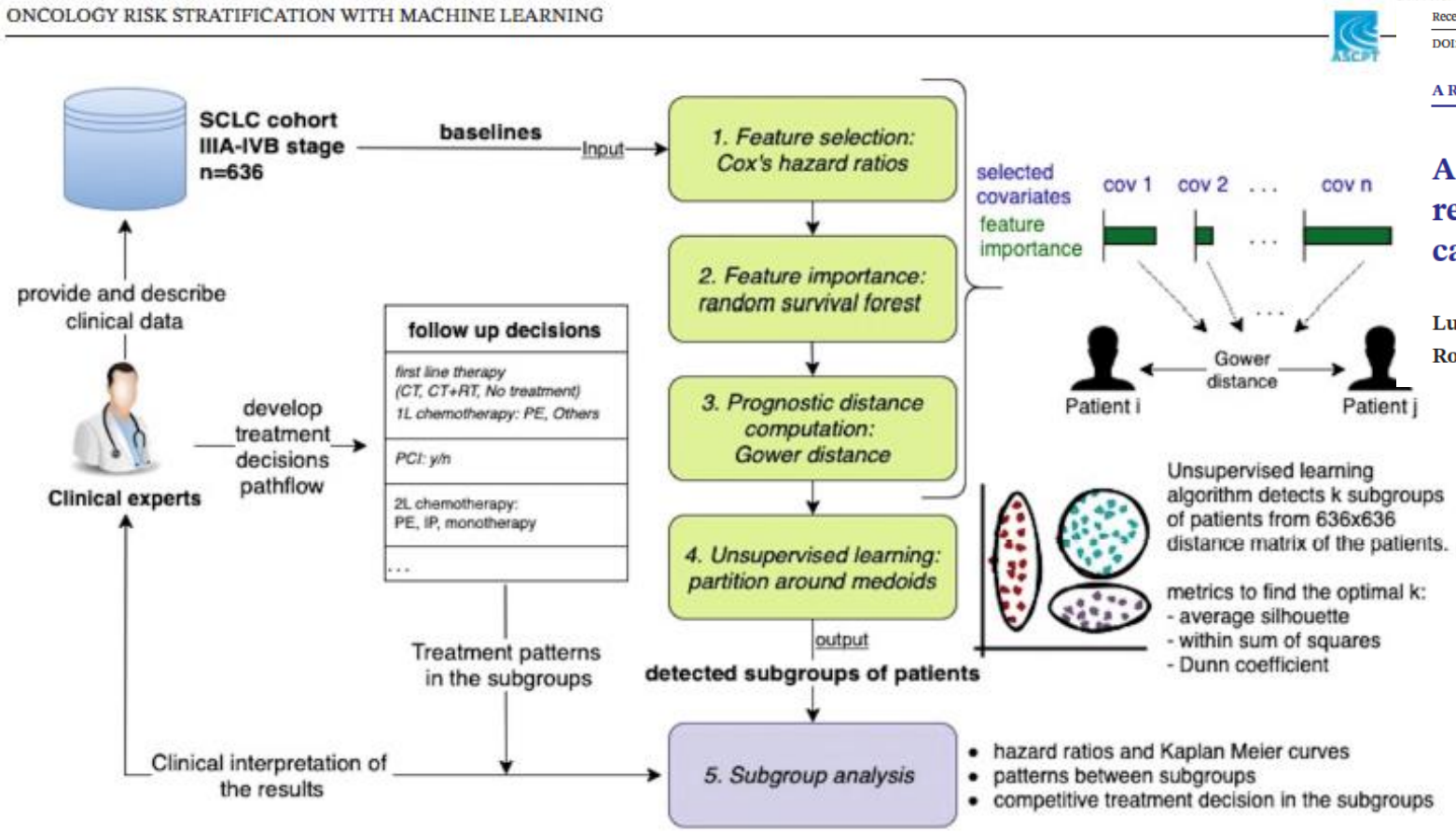


FIGURE 2 Data analysis and model development framework. The aim was to compute a pairwise similarity measure with the most relevant clinical baselines according to their prognosis (steps 1, 2, and 3), group patients with similar characteristics (step 4), and finally, study treatment patterns (Figure 1) of detected groups with thoracic oncologists (step 5). CT, chemotherapy; IP, platinum with irinotecan; PE, platinum with etoposide; RT, radiotherapy; SCLC, small cell lung cancer.

Received: 3 May 2022 | Revised: 26 June 2022 | Accepted: 8 July 2022
DOI: 10.1111/cts.13371

ARTICLE

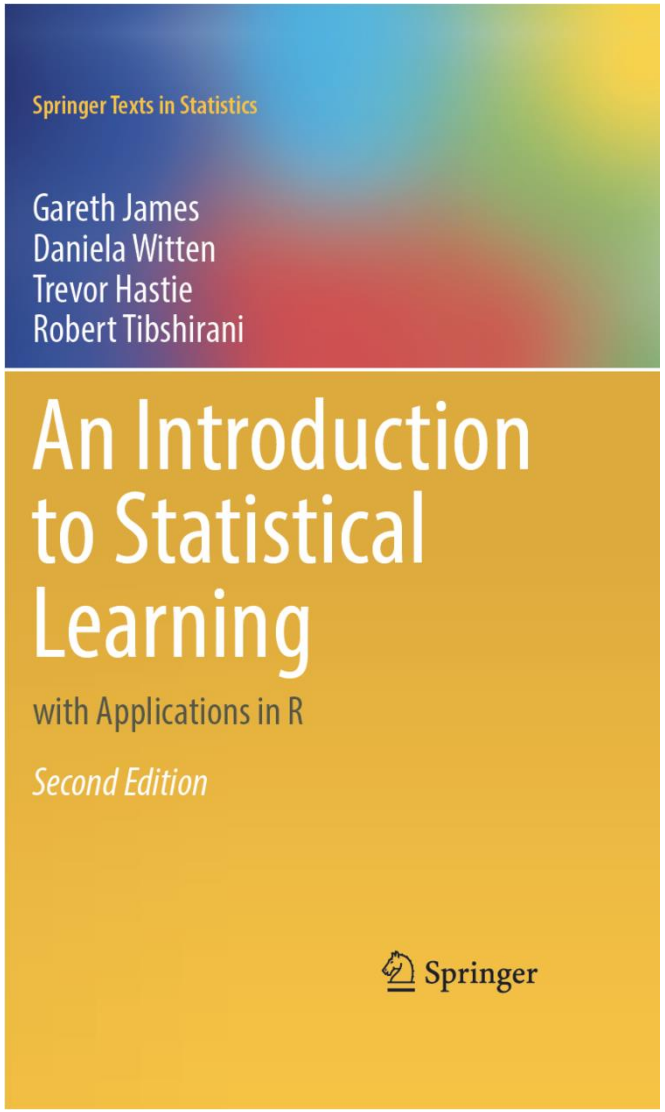
A novel analytical framework for risk stratification of real-world data using machine learning: A small cell lung cancer study

Luca Marzano[1] | Adam S. Darwich[1] | Salomon Tendler[2] | Asaf Dan[2] | Rolf Lewensohn[2] | Luigi De Petris[2] | Jayanth Raghothama[1] | Sebastiaan Meijer[1]

Marzano *et al*. 2022

Statistical learning

Interesting read, not necessary for this course



https://www.statlearning.com